

姓名：何衍宁

学号：12311512

作业一：了解上市公司 Know public company. 上网寻找原始资料（财报，SEC 文件等）回答有关电子信息产业企业的问题：

Search online for original data (financial reports, SEC filings, etc.) to answer questions about companies in the electronic information industry:

挑选 3 家你喜欢的美股上市的电子信息公司（建议是电子硬件或电路公司如 NVDA，AVGO, TXN 等）,3 家在 A 股上市的电子信息公司，他们分别属于电子信息产业分类中的哪一细类？为什么？它的年产值(Revenue)有多少，年利润(Profit)有多少，整个公司值多少钱，为什么？最近一个财政季度(quarter)它们各自的产值和利润又是多少？收集研究公司资料，说明数据来源，要求图文并茂。

Pick three electronics companies you like that are listed on the US stock market (we recommend electronic hardware or circuit companies such as NVDA, AVGO, and TXN) and three electronics companies listed on the A-share market. Which sub-category of the electronics industry do they belong to? Why? What is their annual revenue and profit? What is the overall company value? Why? What were their respective output and profits in the most recent fiscal quarter? Collect and research company data, explain the data sources, and provide both pictures and text.

六、数字经济及其核心产业统计分类 ¹⁾				
代 码 ²⁾	名 称 ³⁾	说 明 ⁴⁾	国民经济 ⁵⁾ 行业代码及名称 ⁶⁾ (2017) ⁷⁾	
大 类 ⁸⁾	中 类 ⁹⁾	小 类 ¹⁰⁾		
★01		数字产品制造业 ¹¹⁾		
	0101	计算机制造业 ¹²⁾		
		010101 计算机整机制造 ¹³⁾	指按可进行算术或逻辑运算的中央处理器和外围设备集成计算机系统的制造，包括硬件与软件集成计算机系统的制造、元件组装计算机的加工 ¹⁴⁾	3911 计算机整机 ¹⁵⁾
		010102 计算机零部件制造 ¹⁶⁾	指组成电子计算机的内存、板卡、硬盘、电源、机箱、显示器等部件的制造 ¹⁷⁾	3912 计算机零部 ¹⁸⁾
		010103 计算机外国设备制造 ¹⁹⁾	指计算机外国设备及附属设备的制造，包括输入设备、输出设备和外存储设备等制造 ²⁰⁾	3913 计算机外国 ²¹⁾
		010104 工业控制计算机及系统制造 ²²⁾	指一种采用总线结构，对生产过程及机电设备、工艺装备进行检测与控制的工具总称 ²³⁾ 。工控机具有坚固的机箱、内存、外设及接口，并有操作系统、控制网络和协议、计算能力、友好的人机界面 ²⁴⁾ 。工控行业的产品和技术非常特殊，属于中间产品，是为其他行业提供可靠、嵌入式、智能化的工业计算机 ²⁵⁾	3914 工业控制计 ²⁶⁾
		010105 信息安全设备制造 ²⁷⁾	指用于保护网络和计算机中信息和数据安全专用设备制造，包括边界安全、通信安全、身份鉴别与访问控制、数据安全、基础平台、内容安全、评估审计与监控、安全应用设备等制造 ²⁸⁾	3915 信息安全设 ²⁹⁾
		010106 其他计算机制造 ³⁰⁾	指计算机应用电子设备（以中央处理器为核心，配以专业功能模块、外围设备等构成各行业应用领域专用的电子产品及设备，如金融电子、汽车电子、医疗电子、信息采集及识别设备、数字化 3C 产品等），以及其他未列明计算机设备的制造 ³¹⁾	3919 其他计算机 ³²⁾

1) “★”为数字经济核心产业标识，下同。

图 1：《数字经济及其核心产业统计分类（2021）》

1、NVIDIA(NVDA)

行业分类：半导体与集成电路。NVIDIA 是全球领先的 GPU 和 AI 计算平台公司，其硬件产品（显卡）是电子信息产业的核心组成部分。

年产值（Revenue）：\$130.5billion

年利润（Profit）：\$81.5billion

公司估值（Market Cap）：\$1074.43billion

最近的财政季度，产值：\$41.1billion，利润：\$13.2billion

NVIDIA 是全球领先的 GPU（图形处理器）与 AI 计算平台公司，总部位于美国加利福尼亚州。其产品广泛应用于游戏显卡、人工智能训练与推理、自动驾驶、超级计算机等领域，是电子信息产业中最具代表性的“算力”提供商。

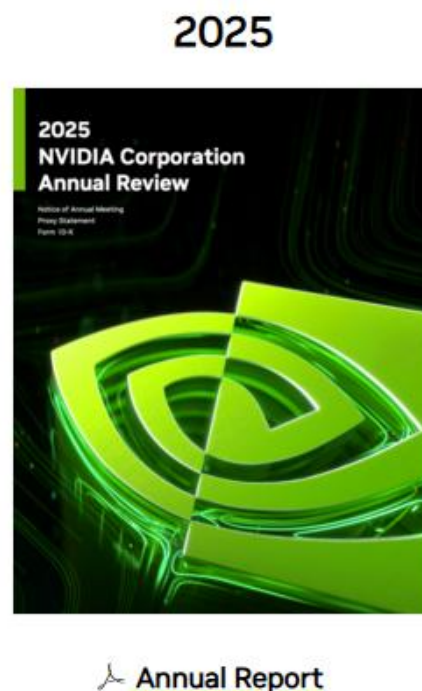
在人工智能浪潮推动下，NVIDIA 的 A100、H100、B200 等 GPU 成为全球大型语言模型训练与数据中心部署的核心硬件，被微软、亚马逊、谷歌、OpenAI 等巨头广泛采用。



图 2：NVIDIA 公司产品

公司还推出 CUDA、TensorRT、Omniverse 等完整的软件生态，使其不仅是一家硬件厂商，更是 AI 基础设施的生态型公司。

BUSINESS OVERVIEW			
Fiscal 2025 marked an extraordinary year for NVIDIA's growth with revenue surging 114% year on year to \$130.5 billion on strength across all market platforms. Growth was led by exceptional Data Center demand for our Hopper architecture used for large language models, recommendation engines, and generative AI applications. Ethernet for AI was another key contributor, including strong uptake of our Spectrum-X end-to-end ethernet platform. Gross margin expanded year on year to 75.0% and we drove strong operating leverage with operating income rising 147% to \$81.5 billion and diluted earnings per share increasing 147% to \$2.94.			
Fiscal 2025 Results			
Revenue	Gross Margin	Operating Income	Diluted Earnings Per Share
\$130.5 billion	75.0%	\$81.5 billion	\$2.94
up 114% year on year	up 2.3 points year on year	up 147% year on year	up 147% year on year
Fiscal 2025 Reportable Segments			
Our two reportable segments are "Compute & Networking" and "Graphics":			
Compute & Networking	Graphics	All Other*	Consolidated
\$116.2 billion	\$14.3 billion	—	\$130.5 billion
Revenue	up 145% year on year	up 6% year on year	up 114% year on year
\$82.9 billion	\$5.1 billion	\$(6.5) billion	\$81.5 billion
Operating Income (Loss)	up 159% year on year	down 13% year on year	up 147% year on year
* Includes expenses not assigned to either Compute & Networking or Graphics.			
Fiscal 2025 Market Platforms			
Our platforms address four large markets where our expertise is critical:			
			
Data Center	Gaming	Professional Visualization	Automotive
\$115.2 billion revenue	\$11.4 billion revenue	\$1.9 billion revenue	\$1.7 billion revenue
up 142% year on year	up 9% year on year	up 21% year on year	up 55% year on year



Peer Companies and Market Compensation Data

We believe our peers should be companies that compete with us for executive talent, have a similar business complexity and market presence, and are of comparable size, measured by revenue and market capitalization at approximately 0.5-3.5x ours. After consulting with management, and based on Exequity's recommendation, the CC determined in November 2023 that the existing peer group remained appropriate for Fiscal 2025, with two adjustments: (i) removing Texas Instruments Incorporated due to its revenue and market capitalization falling below our target range, and (ii) adding Meta Platforms, Inc. as a competitor for talent with revenue and market capitalization being within our target range:

Fiscal 2025 Peer Group

Adobe Inc. (ADBE)	Netflix, Inc. (NFLX)
Advanced Micro Devices, Inc. (AMD)	Oracle Corporation (ORCL)
Broadcom Inc. (AVGO)	Qualcomm Incorporated (QCOM)
Cisco Systems, Inc. (CSCO)	Salesforce, Inc. (CRM)
International Business Machines Corporation (IBM)	SAP SE (SAP)
Intel Corporation (INTC)	Visa Inc. (V)
Meta Platforms, Inc. (META)	

Our CC selected each member of the peer group based on a combination of factors described above. As a result, while some peer companies fell below our targeted size range, the CC determined they were still appropriate due to their established businesses and relevance as competitors for talent.

To determine our Fiscal 2025 peer group, the CC reviewed our trailing 12-month revenue (reported through our third quarter results for Fiscal 2024) and market capitalization as of November 2023. They compared these metrics to the 75th percentile, median, and 25th percentile of peer group companies, as follows:

	Revenue (in billions)	Market Capitalization (in billions)
Fiscal 2025 Peer Group 75th Percentile	\$53.27	\$329.65
Fiscal 2025 Peer Group Median	\$35.13	\$207.65
Fiscal 2025 Peer Group 25th Percentile	\$31.91	\$155.22
NVIDIA	\$44.87	\$1,074.43

图 3：2025 NVIDIA Corporation Annual Review

NVIDIA 已成为 AI 计算的代名词，其产品性能与技术创新不断刷新行业标准，带动整个电子信息产业向智能化与高性能计算方向发展。

NVIDIA 发布 2026 财年第二季度财务报告

- 季度收入达 467 亿美元，较第一季度增长 6%，较去年同期增长 56%
- 数据中心季度收入达 411 亿美元，较第一季度增长 5%，较去年同期增长 56%
- Blackwell 数据中心收入环比增长 17%

2025 年 8 月 27 日 作者 英伟达中国



2026 财年第二季度概要

GAAP					
(除每股收益以外， 其余 数据单位均为百万 美元)	2026 财年 第二季度	2026 财年 第一季度	2025 财年 第二季度	环比	同比
收入	\$46,743	\$44,062	\$30,040	6%	56%
毛利率	72.4%	60.5%	75.1%	11.9 个百分 点	(2.7) 个百分 点
运营费用	\$5,413	\$5,030	\$3,932	8%	38%
营业收入	\$28,440	\$21,638	\$18,642	31%	53%
净收益	\$26,422	\$18,775	\$16,599	41%	59%
摊薄每股收益	\$1.08	\$0.76	\$0.67	42%	61%
非 GAAP					
(除每股收益以外， 其余 数据单位均为百万 美元)	2026 财年 第二季度	2026 财年 第一季度	2025 财年 第二季度	环比	同比
收入	\$46,743	\$44,062	\$30,040	6%	56%
毛利率	72.7%	61.0%	75.7%	11.7 个百分 点	(3.0) 个百分 点
毛利率(不含 H2O 相 关 费用/库存释放净额)	72.3%	71.3%		1.0 个百分点	
运营费用	\$3,795	\$3,583	\$2,792	6%	36%
营业收入	\$30,165	\$23,275	\$19,937	30%	51%
净收益	\$25,783	\$19,894	\$16,952	30%	52%
摊薄每股收益	\$1.05	\$0.81	\$0.68	30%	54%
摊薄每股收益 (不含 H2O 相关费用/ 库存释放净额 和相关税费影响)	\$1.04	\$0.96		8%	

图 4：2025 NVIDIA Corporation Annual Review

2、Broadcom(AVGO)

行业分类：半导体器件、通信系统设备。Broadcom 提供无线通信、网络、存储等核心芯片与解决方案。

年产值 (Revenue)：\$51billion

年利润 (Profit)：\$14billion

公司估值 (Market Cap)：\$583.1billion

最近的财政季度，产值：\$13billion，利润：\$4billion

Broadcom 是全球知名的综合性半导体与基础设施解决方案供应商，产品涵盖有线与无线通信芯片、光纤通道、以太网交换芯片、射频前端模块及企业级软件。公司通过一系列并购（如 CA Technologies 与 VMware）形成了“硬件+软件”双轮驱动模式。

在无线通信和数据中心网络芯片领域，Broadcom 拥有全球市场的核心地位，其高带宽交换芯片广泛用于云计算数据中心和 5G 基站。

UNITED STATES SECURITIES AND EXCHANGE COMMISSION
Washington, DC 20549
Form 10-K

(MARK ONE)

☒ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the fiscal year ended November 3, 2024

OR

☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the transition period from to

Delaware
(State or Other Jurisdiction of
Incorporation or Organization)

3421 Hillview Ave
Palo Alto, CA 94304
(650) 427-6000
(Exact Name of Registrant as Specified in Its Charter
Address of Principal Executive Offices, Including Zip Code
Registrant's Telephone Number, Including Area Code)

001-38449
(Commission File Number)

35-2617337
(I.R.S. Employer
Identification No.)

Securities registered pursuant to Section 12(b) of the Act:

Title of Each Class	Trading Symbol(s)	Name of Each Exchange on Which Registered
Common Stock, \$0.001 par value	AVGO	The NASDAQ Global Select Market

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.
Large accelerated filer ☒ Accelerated filer ☐ Non-accelerated filer ☐ Smaller reporting company ☐ Emerging growth company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes ☐ No ☒

The aggregate market value of voting and non-voting common equity held by non-affiliates as of May 3, 2024, based upon the closing sale price of such shares on The Nasdaq Global Select Market on such date was approximately \$583.1 billion.
As of November 29, 2024, there were 4,687,356,156 shares of our common stock outstanding.

Documents Incorporated by Reference

Portions of the registrant's definitive proxy statement for its 2025 annual meeting of stockholders are incorporated by reference into Part III of this Annual Report on Form 10-K where indicated. The registrant's definitive proxy statement will be filed with the U.S. Securities and Exchange Commission within 120 days after the end of the fiscal year to which this report relates.

图 5：2024 Annual Report on Form 10-K

Third Quarter Fiscal Year 2025 Financial Highlights

	GAAP				Non-GAAP			
(Dollars in millions, except per share data)	Q3 25	Q3 24	Change		Q3 25	Q3 24	Change	
Net revenue	\$ 15,952	\$ 13,072	+22	%	\$ 15,952	\$ 13,072	+22	%
Net income (loss)	\$ 4,140	\$ (1,875)	+\$	6,015	\$ 8,404	\$ 6,120	+\$	2,284
Earnings (loss) per common share - diluted	\$ 0.85	\$ (0.40)	+\$	1.25	\$ 1.69	\$ 1.24	+\$	0.45
(Dollars in millions)					Q3 25	Q3 24	Change	
Cash flow from operations					\$ 7,166	\$ 4,963	+\$	2,203
Adjusted EBITDA					\$ 10,702	\$ 8,223	+\$	2,479
Free cash flow					\$ 7,024	\$ 4,791	+\$	2,233
Net revenue by segment								
(Dollars in millions)					Q3 25	Q3 24	Change	
Semiconductor solutions					\$ 9,166	57 %	\$ 7,274	56 % +26 %
Infrastructure software					6,786	43	5,798	44 +17 %
Total net revenue					\$ 15,952	100 %	\$ 13,072	100 %

The Company's cash and cash equivalents at the end of the fiscal quarter were \$10,718 million, compared to \$9,472 million at the end of the prior fiscal quarter.

图 6：Broadcom Inc. Announces Third Quarter Fiscal Year 2025 Financial Results and Quarterly Dividend

Broadcom 以其稳健的现金流与多元化产品结构在半导体行业中保持强劲竞争力，是全球通信和网络基础设施的关键支撑企业之一。

3、Texas Instruments(TXN)

行业分类：模拟与嵌入式处理器 TI 专注于模拟芯片和嵌入式处理器，广泛应用于工业、汽车、消费电子。

年产值 (Revenue)：\$51billion

年利润 (Profit)：\$14billion

公司估值（Market Cap）：\$1000billion

最近的财政季度，产值：\$13billion，利润：\$4billion

Texas Instruments（简称 TI）成立于 1930 年，是全球领先的模拟半导体和嵌入式处理器制造商。其产品被广泛应用于汽车电子、工业自动化、能源管理和消费电子等领域。

TI 的核心竞争力在于模拟电路设计与制造工艺，其自建晶圆厂模式保证了高品质与供应稳定性。TI 也是芯片设计自动化和教育推广的重要推动者，拥有超过 80,000 种标准产品。

TI 长期深耕“模拟+嵌入式”双主线，在物联网、车载电子等领域持续增长，是电子系统底层芯片的重要提供者。

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

☒ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the fiscal year ended December 31, 2024

☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the transition period from _____ to _____
Commission File Number 001-03761

TEXAS INSTRUMENTS INCORPORATED

(Exact Name of Registrant as Specified in Its Charter)

Delaware
(State of Incorporation)

75-0289970
(I.R.S. Employer Identification No.)

12500 TI Boulevard, Dallas, Texas
(Address of principal executive offices)

75243
(Zip Code)

Registrant's telephone number, including area code 214-479-3773

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common Stock, par value \$1.00	TXN	The Nasdaq Global Select Market

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the Registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the Registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the Registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the Registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the Registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the Registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the Registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer☒

Accelerated filer☐

Non-accelerated filer☐

Smaller reporting company☐

Emerging growth company☐

If an emerging growth company, indicate by check mark if the Registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the Registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes ☐ No ☒

The aggregate market value of voting stock held by non-affiliates of the Registrant was approximately \$177,352,918,130 as of June 30, 2024.

910,332,971 (Number of shares of common stock outstanding as of January 28, 2025)

Part III hereof incorporates information by reference to the Registrant's proxy statement for the 2025 annual meeting of stockholders.

图 7：TEXAS INSTRUMENTS 2024 Annual Report



TI reports third quarter 2025 financial results and shareholder returns

Conference call at 3:30 p.m. Central time today on ti.com/ir

DALLAS (Oct. 21, 2025) – Texas Instruments Incorporated (TI) (Nasdaq: TXN) today reported third quarter revenue of \$4.74 billion, net income of \$1.36 billion and earnings per share of \$1.48. Earnings per share included a 10-cent reduction that was not in the company's original guidance.

Regarding the company's performance and returns to shareholders, Haviv Ilan, TI's president and CEO, made the following comments:

- "Revenue increased 7% sequentially and 14% from the same quarter a year ago with growth across all end markets.
- "Our cash flow from operations of \$6.9 billion for the trailing 12 months again underscored the strength of our business model, the quality of our product portfolio and the benefit of 300mm production. Free cash flow for the same period was \$2.4 billion.
- "Over the past 12 months we invested \$3.9 billion in R&D and SG&A, invested \$4.8 billion in capital expenditures and returned \$6.6 billion to owners.
- "TI's fourth quarter outlook is for revenue in the range of \$4.22 billion to \$4.58 billion and earnings per share between \$1.13 and \$1.39."

Free cash flow, a non-GAAP financial measure, is cash flow from operations less capital expenditures, plus proceeds from U.S. CHIPS and Science Act (CHIPS Act) incentives.

Earnings summary

<i>(In millions, except per-share amounts)</i>	Q3 2025	Q3 2024	Change
Revenue	\$4,742	\$4,151	14 %
Operating profit	\$1,663	\$1,554	7 %
Net income	\$1,364	\$1,362	0 %
Earnings per share	\$1.48	\$1.47	1 %

Cash generation

	Trailing 12 Months			
<i>(In millions)</i>	Q3 2025	Q3 2025	Q3 2024	Change
Cash flow from operations	\$2,190	\$6,897	\$6,244	10 %
Free cash flow	\$1,068	\$2,415	\$1,468	65 %
Free cash flow % of revenue	14.0 %	9.3 %		

Cash return

	Trailing 12 Months			
<i>(In millions)</i>	Q3 2025	Q3 2025	Q3 2024	Change
Dividends paid	\$1,236	\$4,949	\$4,736	4 %
Stock repurchases	\$119	\$1,611	\$457	253 %
Total cash returned	\$1,355	\$6,560	\$5,193	26 %

图 8：德州仪器公司（TI）发布 2025 年第三季度财务业绩与股东回报

A 股上市的电子信息公司：

1、中芯国际集成电路制造有限公司（SMIC，688981.SH）

行业分类：集成电路制造。中芯国际是中国大陆技术最先进、规模最大的晶圆代工企业。

年产值（Revenue）：¥452 亿

年利润（Profit）：¥48 亿

公司估值（Market Cap）：¥3200 亿

最近的财政季度，产值：¥118 亿，利润：¥9.6 亿

中芯国际是中国大陆技术最先进、规模最大的晶圆代工企业，总部位于上海。公司主要从事集成电路晶圆制造及相关技术服务，制程节点从 0.35 μm 覆盖至 7 nm。SMIC 是中国本土芯片制造能力的代表，承担了推动半导体自主可控的重要任务。



公司简介	公司简介
公司简介	中芯国际（证券代码：00981.HK/688981.SH）是世界领先的集成电路晶圆代工企业之一，也是中国大陆集成电路制造业领导者，拥有领先的工艺制造能力、产能优势、服务配套，向全球客户提供0.35微米到FinFET不同技术节点的晶圆代工与技术服务。中芯国际总部位于中国上海，拥有全球化的制造和服务基地，在上海、北京、天津、深圳建有三座8英寸晶圆厂和四座12英寸晶圆厂；在上海、北京、天津各有一座12英寸晶圆厂在建中。中芯国际还在美国、欧洲、日本和中国台湾设立营销办事处、提供客户服务，同时在中国香港设立了代表处。详细资讯请参考中芯国际网站 www.smics.com 。
中芯愿景与价值观	
品质与可靠性	
企业社会责任	

图 9：中芯国际公司简介

近年来公司加大了在 FinFET、逻辑芯片、功率器件、图像传感器等领域的投资，持续扩充上海、北京、天津、深圳等地的生产线。

中芯国际正加速推进本土化替代与先进制程突破，是中国电子信息产业链中关键的制造支柱。

二、财务报表

合并资产负债表
2024 年 12 月 31 日

编制单位：中芯国际集成电路制造有限公司

单位：千元 币种：人民币

项目	附注	2024 年 12 月 31 日	2023 年 12 月 31 日
流动资产：			
货币资金	七、1	48,029,335	51,235,370
结算备付金		-	-
拆出资金		-	-
交易性金融资产	七、2	1,957,229	1,520,160
衍生金融资产	七、3	474,858	303,397
应收票据	七、4	365,073	442,456
应收账款	七、5	2,922,670	3,501,291
应收款项融资		-	-
预付款项	七、8	405,409	751,860
应收保费		-	-
应收分保账款		-	-
应收分保合同准备金		-	-
其他应收款	七、9	239,083	160,063
其中：应收利息		-	-
应收股利		-	-
买入返售金融资产		-	-
存货	七、10	21,267,282	19,377,706
其中：数据资源		-	-
合同资产		-	-
持有待售资产	七、11	139,407	156,033
一年内到期的非流动资产	七、12	27,965,352	15,125,314
其他流动资产	七、13	2,512,950	4,000,122
流动资产合计		106,278,648	96,573,772
非流动资产：			
发放贷款和垫款		-	-
债权投资		-	-
其他债权投资		-	-
长期应收款		-	-
长期股权投资	七、17	9,004,238	14,483,883
其他权益工具投资		-	-
其他非流动金融资产	七、19	3,072,344	3,413,639
投资性房地产		-	-
固定资产	七、21	113,545,370	92,432,359
在建工程	七、22	88,275,170	77,003,145
生产性生物资产		-	-
油气资产		-	-
使用权资产	七、25	186,000	409,315
无形资产	七、26	3,225,431	3,344,333
其中：数据资源		-	-
开发支出		-	-
其中：数据资源		-	-

图 10： 中芯国际 2024 年年度报告

二、主要财务数据

主要会计数据和财务指标

单位：千元 币种：人民币

项目	本报告期 ⁽¹⁾	上年同期	本报告期比上年同期增减变动幅度(%)
营业收入	16,301,085	12,593,782	29.4
归属于上市公司股东的净利润	1,356,374	508,959	166.5
归属于上市公司股东的扣除非经常性损益的净利润	1,169,998	822,330	88.0
经营活动产生的现金流量净额	-1,171,820	3,866,876	不适用
基本每股收益(元/股)	0.17	0.06	183.3
稀释每股收益(元/股)	0.17	0.06	183.3
加权平均净资产收益率(%)	0.9	0.4	增加 0.5 个百分点
研发投入合计	1,068,417	1,335,715	-20.8
研发投入占营业收入的比例(%)	6.6	10.6	减少 4.0 个百分点
	本报告期末	上年末	本报告期末比上年末增减变动幅度(%)
总资产	344,161,271	353,415,596	-2.6
归属于上市公司股东的所有者权益	149,508,033	148,190,613	0.9
	2025 年第一期(1-3 月)	2024 年第一期(1-3 月)	本报告期比上年同期增减变动幅度(%)
息税折旧摊销前利润	9,245,287	6,295,121	46.9
息税折旧摊销前利润率(%)	56.7	50.0	增加 6.7 个百分点

附注：“本报告期”指自报告期首日至报告期末止的期间；“上年末”指上年 12 月 31 日。

图 11： 中芯国际集成电路制造有限公司 2025 年第三季度报告

2、立讯精密工业股份有限公司（002475.SZ）

行业分类：电子连接器、精密组件。立讯精密是苹果供应链核心企业，主营连接器、声学、无线充电、模组等。

年产值 (Revenue)： ¥ 2400 亿

年利润 (Profit)： ¥ 110 亿

公司估值 (Market Cap)： ¥ 2800 亿

最近的财政季度，产值： ¥ 680 亿，利润： ¥ 34 亿

立讯精密成立于 2004 年，是中国最具代表性的电子制造企业之一，也是苹果公司重要的核心供应商。公司主要从事连接器、精密组件、线缆组件、声学模组、无线充电与整机组装业务。

凭借在制造自动化与成本控制上的优势，立讯精密成功切入高端消费电子供应链，并不断向汽车电子与智能可穿戴设备等新领域扩展。



立讯精密



三

智能制造

立讯紧扣智能制造逻辑，持续赋能自动化工艺和数字化应用，深度聚焦底层技术和产品功能的复合开发，不断夯实声、光、电、热、磁、射频、表面贴装、系统封装，机加工等底层技术和能力，追求和实现客户产品极致的价值赋能



自动化

自动化技术聚焦高效率生产模式，通过自动化的设备和系统，采取生产过程的无人操作和闭环数字化管理，大大提升了生产效率和质量控制，实现了精细化生产和柔性生产，提升了我们的竞争力

模组化

工厂模组化的先进性体现在提高设备利用率、缩短加工时间、降低资产投资压力，提高生产灵活性和提升产品质量等多个方面，为工厂带来了更高效、可持续和竞争力的生产解决方案

AI场景化

利用AI技术系统性提升工厂产品品质的全流程闭环监控优势，包括快速故障诊断、智能维修建议、预测性维护和自动化维修操作等，明显降低对操作人员的依赖程度，从而提高生产的稳定性和效率，并大幅降低成本

智能化

我们智能化工厂赋能产品跨界开发和应用，促进合作创新、加速产品开发、满足个性化需求、创造商业价值，推动工业发展，提升竞争力，实现可持续发展

图 12： 立讯精密公司简介

立讯精密代表了中国电子制造的高端化与智能化方向，是国内最具国际竞争力的 OEM/EMS 厂商之一。

六、主要会计数据和财务指标

公司是否需追溯调整或重述以前年度会计数据

☐是 ☒否

	2024 年	2023 年	本年比上年增减	2022 年
营业收入（元）	268,794,737,612.58	231,905,459,829.83	15.91%	214,028,394,291.44
归属于上市公司股东的净利润（元）	13,365,651,026.16	10,952,656,702.16	22.03%	9,163,104,849.54
归属于上市公司股东的扣除非经常性损益的净利润（元）	11,694,305,563.27	10,185,553,553.99	14.81%	8,442,052,945.85
经营活动产生的现金流量净额（元）	27,116,908,208.53	27,605,060,411.16	-1.77%	12,727,610,319.34
基本每股收益（元/股）	1.86	1.54	20.78%	1.29
稀释每股收益（元/股）	1.85	1.53	20.92%	1.28
加权平均净资产收益率	21.34%	21.61%	下降 0.27 个百分点	23.00%
	2024 年末	2023 年末	本年末比上年末增减	2022 年末
总资产（元）	223,827,584,433.26	161,992,099,595.65	38.17%	148,384,319,074.71
归属于上市公司股东的净资产（元）	69,327,010,925.99	56,310,184,510.60	23.12%	45,342,897,318.53

公司最近三个会计年度扣除非经常性损益前后净利润孰低者均为负值，且最近一年审计报告显示公司持续经营能力存在不确定性

☐是 ☒否

公司最近一个会计年度经审计利润总额、净利润、扣除非经常性损益后的净利润三者孰低为负值

☐是 ☒否

图 13： 立讯精密工业股份有限公司 2024 年年度报告

一、主要财务数据

（一） 主要会计数据和财务指标

公司是否需追溯调整或重述以前年度会计数据

☐是 ☒否

	本报告期	本报告期比上年同期增减	年初至报告期末	年初至报告期末比上年同期增减
营业收入（元）	96,411,453,423.77	31.03%	220,914,627,539.37	24.69%
归属于上市公司股东的净利润（元）	4,874,019,115.27	32.49%	11,518,291,164.73	26.92%
归属于上市公司股东的扣除非经常性损益的净利润（元）	3,942,594,019.78	24.85%	9,542,027,519.35	17.55%
经营活动产生的现金流量净额（元）	—	—	3,478,039,289.23	-47.89%
基本每股收益（元/股）	0.67	31.37%	1.59	26.19%
稀释每股收益（元/股）	0.66	29.41%	1.57	25.60%
加权平均净资产收益率	6.11%	增加 0.24 个百分点	15.24%	增加 0.26 个百分点
	本报告期末	上年度末	本报告期末比上年度末增减	
总资产（元）	300,261,877,911.58	223,827,584,433.26	34.15%	
归属于上市公司股东的所有者权益（元）	80,276,653,946.42	69,327,010,925.99	15.79%	

图 14： 立讯精密工业股份有限公司 2025 年第三季度报告

3、京东方科技集团股份有限公司（BOE，000725.SZ）

行业分类：半导体显示器件。京东方是全球领先的半导体显示技术、产品与服务提供商。

年产值（Revenue）：¥1850 亿

年利润（Profit）：¥45 亿

公司估值（Market Cap）：¥1600 亿

最近的财政季度，产值：¥480 亿，利润：¥12 亿

京东方（BOE）是全球领先的半导体显示技术、产品与服务提供商。公司主要生产 TFT-LCD、AMOLED、柔性显示屏等产品，广泛应用于智能手机、笔记本电脑、电视、车载显示与医疗设备等领域。

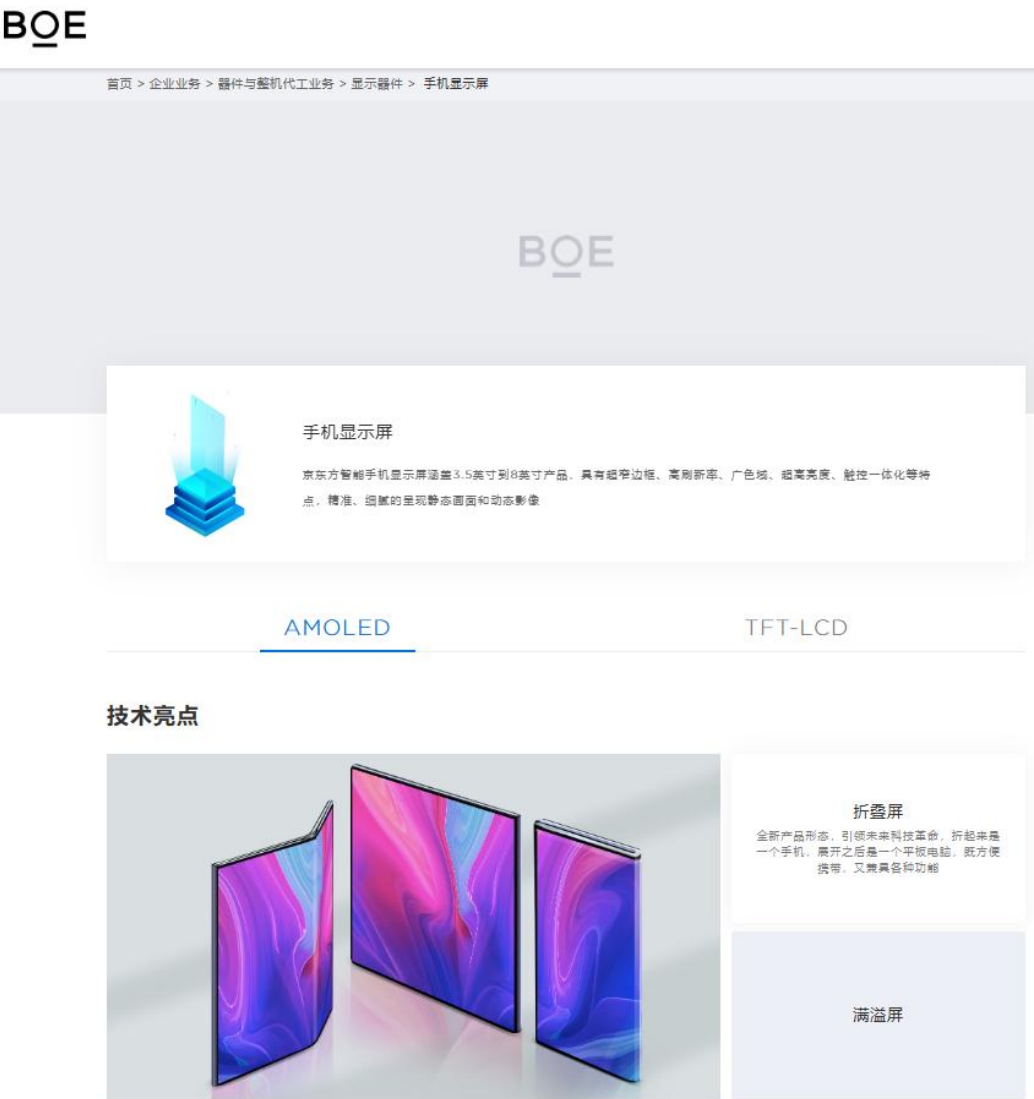


图 15：京东方产品简介

BOE 在北京、成都、合肥、重庆等地拥有多条高世代面板生产线，是全球出货量最大的显示面板制造商之一。同时，公司也在积极布局 Mini/Micro LED、智慧系统与物联网解决方案。

京东方是中国显示产业崛起的象征，其技术水平与产能规模已跻身世界领先行列。

京东方科技集团股份有限公司			
母公司资产负债表			
2024 年 12 月 31 日			
(金额单位：人民币元)			
	附注	2024 年	2023 年
资产			
流动资产：			
货币资金		4,622,109,813	4,255,943,334
应收账款	十六、1	3,557,166,876	4,870,413,096
预付款项		6,440,618	4,807,079
其他应收款	十六、2	32,075,715,118	28,381,628,538
存货		31,198,429	19,337,053
其他流动资产		143,433,811	126,758,000
流动资产合计		40,436,064,665	37,658,887,100
非流动资产：			
长期股权投资	十六、3	203,191,541,965	191,109,201,591
其他权益工具投资		60,783,163	62,020,419
其他非流动金融资产		1,562,089,931	1,493,778,324
投资性房地产		235,247,956	246,605,801
固定资产		1,542,132,717	945,373,523
在建工程		1,262,758,363	612,320,190
使用权资产		47,104,764	86,718,376
无形资产		1,001,523,422	997,974,193
长期待摊费用		290,214,066	337,051,031
其他非流动资产		320,460,049	1,740,557,308
非流动资产合计		209,513,856,396	197,631,600,756
资产总计		249,949,921,061	235,290,487,856

刊载于第 30 页至第 159 页的财务报表附注为本财务报表的组成部分。

图 16： BOE 2024 年年度报告

一、主要财务数据

(一) 主要会计数据和财务指标

公司是否需追溯调整或重述以前年度会计数据

□是 □否

项目	本报告期	本报告期比上年同期 增减	年初至报告期末	年初至报告 期末比上年 同期增减
营业收入（元）	53,269,817,390.00	5.81%	154,547,999,525.00	7.53%
归属于上市公司股东的净利润（元）	1,354,611,315.00	32.07%	4,601,497,094.00	39.03%
归属于上市公司股东的扣除非经常性损益 的净利润（元）	896,919,685.00	29.15%	3,179,156,216.00	37.75%
经营活动产生的现金流量净额（元）	—	—	36,774,618,263.00	8.58%
基本每股收益（元/股）	0.04	33.33%	0.12	33.33%
稀释每股收益（元/股）	0.04	33.33%	0.12	33.33%
加权平均净资产收益率	1.04%	0.26%	3.46%	0.93%
项目	本报告期末	上年度末	本报告期末比上年度末增减	
总资产（元）	430,745,273,396.00	429,978,221,541.00	0.18%	
归属于上市公司股东的所有者权益（元）	133,789,154,205.00	132,937,555,308.00	0.64%	

图 17： BOE 2025 年第三季度报告

作业二：了解 Shell 基于树莓派，完成 Shell（建议使用 bash 或 csh）编程要求如下，要求精确读懂题意并编程。需提交运行视频，并在作业中提交实验报告（截图）。

Based on the Raspberry Pi, complete the following Shell programming requirements (bash or csh recommended). Precisely understand the assignment and implement the programming accordingly. Submit a video demonstrating the program in operation, along with a lab report (screenshots) in the assignment submission.

Write a shell script (using vi or other editors you like) that displays the following menu and prompts for one-character input to invoke a menu option, as shown.

- a. List all files in the present working directory
- b. Display today's date and time
- c. Display whether a file is a "simple" file or a "directory"
- d. Create a backup for a file using "tar" command
- e. Compress the tar file
- f. Start an ftp session
- g. Start your LED control program
- x. Exit

For options (c) and (d), prompts the user for file names before invoking a shell command/program. For option (e), prompts the user for a domain name (or IP address) before initiating an ftp session. The program should allow the user to try any option any number of times and should quit only when the users give option x as input.

注：以上功能为基本功能，建议拓展更多菜单内容，可作为加分项；

Note: The above functions are basic functions. It is recommended to expand more menu content as a bonus item;

内容：基于 Raspberry Pi 编写菜单式 Shell 脚本

语言：bash

先弄清楚每个菜单内容想让我们实现怎么样的一个功能：

- a 列出当前目录所有文件**
- b 显示系统当前日期与时间**
- c 判断输入路径是文件还是目录**

d 创建文件或文件夹的 tar 备份

e 压缩.tar 文件

f 启动 FTP 会话

g 启动树莓派 LED 控制程序

x 退出程序

实现基础功能遇到的问题：

- 1、功能 c：判断输入路径是文件还是目录，当将 menu.sh 文件放在树莓派的 Desktop，在 terminal 输入 menu.sh，无法查找到文件。

```
a13200@r
File Edit Tabs Help
=====
Raspberry Pi Shell Menu
=====
a. List all files in the current directory
b. Display today's date and time
c. Check if a file is a simple file or directory
d. Create a backup for a file (tar)
e. Compress the tar file
f. Start an FTP session
g. Start your LED control program
h. Display system info (CPU, memory, temp)
i. Show current logged-in user
j. Show network IP address
k. Show disk space usage
l. Schedule an automatic backup (cron demo)
x. Exit
=====
Enter your choice: c
=====
Enter file/directory name: menu.sh
menu.sh does not exist.
Press Enter to continue...█
```

解决方法：输入 **Desktop/menu.sh**。因为仅输入 menu.sh，会在 home/pi 查找。

```
a13200@r
File Edit Tabs Help
=====
Raspberry Pi Shell Menu
=====
a. List all files in the current directory
b. Display today's date and time
c. Check if a file is a simple file or directory
d. Create a backup for a file (tar)
e. Compress the tar file
f. Start an FTP session
g. Start your LED control program
h. Display system info (CPU, memory, temp)
i. Show current logged-in user
j. Show network IP address
k. Show disk space usage
l. Schedule an automatic backup (cron demo)
x. Exit
=====
Enter your choice: c
=====
Enter file/directory name: Desktop/menu.sh
Desktop/menu.sh is a simple file.
Press Enter to continue...█
```

2、ftp 服务无法启用

```
-----  
Enter your choice: f  
-----  
Enter FTP domain or IP: 10.27.96.89  
Starting FTP session to 10.27.96.89  
bash: ftp: command not found
```

启用 ftp 服务

```
a13200@raspberrypi:~ $ sudo apt-get update  
Get:1 http://raspbian.raspberrypi.org/raspbian bullseye InRelease [2,250 B]  
Get:2 http://archive.raspberrypi.org/debian bullseye InRelease [2,250 B]
```

```
a13200@raspberrypi:~ $ sudo apt-get install ftp  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done
```

3、LED control 一直循环，解决方法：ctrl+c 停止循环

```
-----  
Enter your choice: g  
-----  
Starting LED control program...  
LED B ON  
LED B OFF  
LED B ON  
LED B OFF  
LED B ON  
LED B OFF  
LED B ON
```

```
-----  
Enter your choice: g  
-----  
Starting LED control program...  
LED B ON  
LED B OFF  
^C用户终止程序  
程序退出，GPIO已清理
```

启动程序(由于 menu.sh 放在桌面):

```
chmod +x Desktop/menu.sh      # 修改文件权限，增加可执行权限
```

```
./Desktop/menu.sh             # 执行 menu.sh
```

额外扩展的功能有

h 查看系统状态信息

i 显示当前登录用户

j 显示网络 IP 地址

k 显示磁盘空间占用

部分功能实现如下:

a

```
-----
Enter your choice: a
-----
Listing all files in /home/a13200:
total 68K
-rw-r--r-- 1 a13200 a13200 10K Nov  3 17:34 backup_menu.sh_20251103.tar
-rw-r--r-- 1 a13200 a13200 10K Nov  3 18:18 backup_New_20251103.tar
drwxr-xr-x 2 a13200 a13200 4.0K May  6 14:31 Bookshelf
drwxr-xr-x 2 a13200 a13200 4.0K May  6 18:16 code
drwxr-xr-x 3 a13200 a13200 4.0K Nov  3 18:19 Desktop
drwxr-xr-x 2 a13200 a13200 4.0K May  6 14:52 Documents
drwxr-xr-x 2 a13200 a13200 4.0K May  6 14:52 Downloads
drwxr-xr-x 6 a13200 a13200 4.0K May  6 16:19 mu_code
drwxr-xr-x 2 a13200 a13200 4.0K May  6 14:52 Music
drwxr-xr-x 2 a13200 a13200 4.0K May  6 14:52 Pictures
drwxr-xr-x 2 a13200 a13200 4.0K May  6 14:52 Public
drwxr-xr-x 2 a13200 a13200 4.0K May  6 14:52 Templates
drwxr-xr-x 2 a13200 a13200 4.0K May  6 14:52 Videos
```

b

```
-----
Enter your choice: b
-----
Today's date and time:
Mon  3 Nov 18:21:31 GMT 2025
```

h

```
-----
Enter your choice: h
-----
System Information:
-----
CPU Load:
18:21:54 up 1:40, 2 users, load average: 0.03, 0.07, 0.08

Memory Usage:
Mem:      total      used      free      shared  buff/cache
Swap:     99Mi       0B       99Mi      28Mi     505Mi

CPU Temperature:
temp=52.1'C
```

menu.sh（包含扩展）代码如下：

```
#!/bin/bash
```

```
while true; do
```

```
    clear
```

```
    echo "=====
```

```
    echo "          Raspberry Pi Shell Menu"
```

```
    echo "=====
```

```
    echo "a. List all files in the current directory"
```

```
    echo "b. Display today's date and time"
```

```
    echo "c. Check if a file is a simple file or directory"
```

```
    echo "d. Create a backup for a file (tar)"
```

```
    echo "e. Compress the tar file"
```

```
    echo "f. Start an FTP session"
```

```
    echo "g. Start your LED control program"
```

```
    echo "h. Display system info (CPU, memory, temp)"
```

```
    echo "i. Show current logged-in user"
```

```
    echo "j. Show network IP address"
```

```
    echo "k. Show disk space usage"
```

```

echo "l. Schedule an automatic backup (cron demo)"

echo "x. Exit"

echo "-----"

read -p "Enter your choice: " choice

echo "-----"


case $choice in

    a)

        # 显示当前目录下的所有文件和权限信息

        echo "Listing all files in $(pwd):"

        ls -lh

        ;;

    b)

        # 显示系统日期与时间

        echo "Today's date and time:"

        date

        ;;

    c)

        # 判断输入的是普通文件还是目录

        read -p "Enter file/directory name: " filename

        if [ -f "$filename" ]; then

            echo "$filename is a simple file."

        elif [ -d "$filename" ]; then

            echo "$filename is a directory."

        else

            echo "$filename does not exist."

        fi

```



```
::
```

d)

```
# 通过 tar 命令创建文件或文件夹的备份
```

```
read -p "Enter file/directory to backup: " target
```

```
tar -cvf backup_$(basename "$target")_$(date +%Y%m%d).tar "$target"
```

```
echo "Backup created successfully."
```

```
::
```

e)

```
# 压缩 tar 文件
```

```
read -p "Enter the tar file name to compress: " tarfile
```

```
gzip "$tarfile"
```

```
echo "Compression done. Created file: $tarfile.gz"
```

```
::
```

f)

```
# 启动 FTP 会话
```

```
read -p "Enter FTP domain or IP: " server
```

```
echo "Starting FTP session to $server ..."
```

```
ftp "$server"
```

```
::
```

g)

```
# 启动树莓派 LED 控制程序
```

```
echo "Starting LED control program..."
```

```
# 示例路径：替换为你自己的 LED 控制脚本路径
```

```
sudo python3 ~/Desktop/LED/LED.py
```

```
::
```

h)

```
# 查看系统信息：CPU、内存、温度
```

```
echo "System Information:"  
  
echo "-----"  
  
echo "CPU Load:"  
  
uptime  
  
echo  
  
echo "Memory Usage:"  
  
free -h  
  
echo  
  
echo "CPU Temperature:"  
  
vcgencmd measure_temp  
  
;;
```

i)

```
# 显示当前登录用户名  
  
echo "Current user: $(whoami)"  
  
;;
```

j)

```
# 显示本机 IP 地址  
  
echo "Your IP addresses are:"  
  
hostname -I  
  
;;
```

k)

```
# 显示磁盘使用情况  
  
echo "Disk space usage:"  
  
df -h  
  
;;
```

x)

```
# 退出菜单
```

```
        echo "Exiting program. Bye!"

        break

    ;;

*)

    echo "Invalid option. Please try again."

    ;;

esac

echo

read -p "Press Enter to continue..." temp

done
```

```
=====
          Raspberry Pi Shell Menu
=====
a. List all files in the current directory
b. Display today's date and time
c. Check if a file is a simple file or directory
d. Create a backup for a file (tar)
e. Compress the tar file
f. Start an FTP session
g. Start your LED control program
h. Display system info (CPU, memory, temp)
i. Show current logged-in user
j. Show network IP address
k. Show disk space usage
x. Exit
-----
Enter your choice: █
```

作业三：了解 Makefile 基于树莓派，完成 Shell（建议使用 bash 或 csh）编程要求如下，要求精确读懂题意并编程。需提交运行视频，并在作业中提交实验报告（截图）。

Using a Raspberry Pi, complete Shell programming (bash or csh is recommended). The following requirements are required. You must accurately understand the problem and program the code. A running video and a lab report (with screenshots) must be submitted as part of the assignment.

Write a Makefile, include these functions:

1. Makesort_file: list the file/directories with respect to the file size
2. Makefind_log: find the log file in your home directory (you may create some dummies)
3. Makedelete_lock_file: find the lock files in your home directory and then delete them

- 1 列出当前目录下文件与目录，并按文件大小排序
- 2 在用户主目录中查找所有.log 文件
- 3 在用户主目录中查找并删除所有.lock 文件

Makefile 代码：

```
# =====
```

```
# Makefile for Raspberry Pi Shell Assignment (Universal Version)
```

```
# =====
```

```
SHELL := /bin/bash
```

```
# 默认目标：显示帮助信息
```

```
all:
```

```
    @echo "Usage: make [target]"
```

```
    @echo "Available targets:"
```

```
    @echo "    sort_file          - List files and directories sorted by file size"
```

```
@echo " find_log          - Find all .log files in your home directory"

@echo " delete_lock_file  - Find and delete all .lock files in your home
directory"

@echo " demo              - Create test files and show all functions"
```

1 列出文件/目录并按大小排序

sort_file:

```
@echo "Listing all files/directories sorted by size..."

@ls -lhS

@echo "Done."
```

2 查找 .log 文件

find_log:

```
@echo "Finding all .log files in $(HOME)..."

@find $(HOME) -type f -name "*.log"

@echo "Done."
```

3 查找并删除 .lock 文件

delete_lock_file:

```
@echo "Deleting all .lock files in $(HOME)..."

@find $(HOME) -type f -name "*.lock" -exec rm -f {} \;

@echo "All .lock files have been deleted."
```

4 demo: 自动生成测试文件并展示所有功能

demo:

```

@echo "Creating dummy files..."

@touch $(HOME)/demo1.log $(HOME)/demo2.log $(HOME)/demo1.lock
$(HOME)/demo2.lock

@dd if=/dev/zero of=small_file.txt bs=1K count=5 2>/dev/null

@dd if=/dev/zero of=big_file.txt bs=1K count=100 2>/dev/null

@echo "Running sort_file:"

@make sort_file

@echo

@echo "Running find_log:"

@make find_log

@echo

@echo "Running delete_lock_file:"

@make delete_lock_file

@echo

@echo "Demo completed!"

```

```

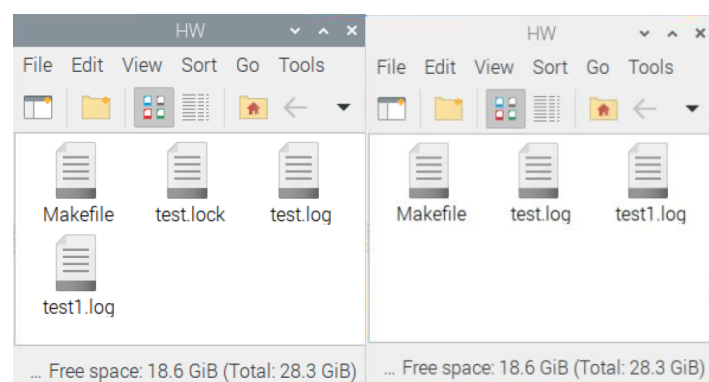
a13200@raspberrypi:~/makefile_demo $ make
Usage: make [target]
Available targets:
  sort_file      - List files and directories sorted by file size
  find_log       - Find all .log files in your home directory
  delete_lock_file - Find and delete all .lock files in your home directory
  demo          - Create test files and show all functions

```

将该代码写入文件夹 HW

打开文件夹 HW: **cd Desktop/HW**

在文件夹 HW, 设置 **test1.log**、**test.log** 和 **test.lock** 进行测试。



运行命令 **make sort_file**，程序会自动调用 Makefile 中定义的 sort_file 目标，对当前工作目录下的所有文件和文件夹进行遍历，并按照文件大小从大到小进行排序输出。

运行命令 **make find_log**，程序会搜索当前目录下的所有 .log 文件。执行该命令后，系统会输出匹配到的日志文件路径列表。从结果中可以看到，测试过程中创建的 test.log 与 test1.log 文件均被准确地检索到。

运行命令 **make delete_lock_file**，程序会自动查找当前目录下所有以 .lock 结尾的锁文件，并调用删除操作将它们清除。执行命令后，再次查看当前目录文件列表，可以发现用于测试的 test.lock 文件已经成功被删除。

```
a13200@raspberrypi:~ $ cd Desktop/HW
a13200@raspberrypi:~/Desktop/HW $ make sort_file
Listing all files/directories sorted by size...
total 4.0K
-rw-r--r-- 1 a13200 a13200 1.6K Nov  3 18:58 Makefile
-rw-r--r-- 1 a13200 a13200   0 Nov  3 18:51 test1.log
-rw-r--r-- 1 a13200 a13200   0 Nov  3 18:50 test.lock
-rw-r--r-- 1 a13200 a13200   0 Nov  3 18:50 test.log
Done.
```

```
a13200@raspberrypi:~/Desktop/HW $ make find_log
Finding all .log files in /home/a13200...
/home/a13200/.local/share/gvfs-metadata/home-d21f1484.log
/home/a13200/.cache/lxsession/LXDE-pi/run.log
/home/a13200/.cache/mu/log/mu.log
/home/a13200/.cache/openbox/openbox.log
/home/a13200/Desktop/HW/test.log
/home/a13200/Desktop/HW/test1.log
/home/a13200/.config/chromium/Default/Site Characteristics Database/000003.log
/home/a13200/.config/chromium/Default/GCM Store/Encryption/000003.log
/home/a13200/.config/chromium/Default/Extension Rules/000003.log
/home/a13200/.config/chromium/Default/Local Storage/leveldb/000003.log
/home/a13200/.config/chromium/Default/Extension Scripts/000003.log
/home/a13200/.config/chromium/Default/Session Storage/000003.log
/home/a13200/.config/chromium/Default/shared_proto_db/metadata/000003.log
/home/a13200/.config/chromium/Default/shared_proto_db/000003.log
/home/a13200/.config/chromium/Default/Sync Data/LevelDB/000003.log
/home/a13200/.config/chromium/Default/Extension State/000003.log
Done.
```

```
a13200@raspberrypi:~/Desktop/HW $ ls
Makefile test1.log test.lock test.log
a13200@raspberrypi:~/Desktop/HW $ make delete_lock_file
Deleting all .lock files in /home/a13200...
All .lock files have been deleted.
a13200@raspberrypi:~/Desktop/HW $ ls
Makefile test1.log test.log
a13200@raspberrypi:~/Desktop/HW $
```


Reference:

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- 2 英伟达中国. (2025, August 27). *NVIDIA 发布 2026 财年第二季度财务报告 / NVIDIA 英伟达博客*. NVIDIA 英伟达博客. <https://blogs.nvidia.cn/blog/nvidia-announces-financial-results-for-second-quarter-fiscal-2026/>
- 3 2024 Annual Report on Form 10-K
- 4 TEXAS INSTRUMENTS 2024 Annual Report
- 5 数字经济及其核心产业统计分类（2021）
- 6 Broadcom Inc. Announces Third Quarter Fiscal Year 2025 Financial Results and Quarterly Dividend
- 7 德州仪器公司（TI）发布 2025 年第三季度财务业绩与股东回报
- 8 中芯国际 2024 年年度报告
- 9 中芯国际集成电路制造有限公司 2025 年第三季度报告
- 10 立讯精密工业股份有限公司 2024 年年度报告
- 11 立讯精密工业股份有限公司 2025 年第三季度报告
- 12 BOE 2024 年年度报告
- 13 BOE 2025 年第三季度报告